

CS & IT ENGINEERING



Computer Network - 2

Transport & Application Layer

Discussion Notes

DPP - 01



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[MSQ]

#Q. Which of the following socket API function is/are executed by TCP Server ?

☒ A

bind()

☐ B

connect()

→ TCP Client

☒ C

listen()

☒ D

accept()

TCP SERVER

Socket()

Bind()

Listen()

Accept()

Ans: A, C & D

[NAT]



#Q. Consider a TCP connection with end-to-end transmission bandwidth is 262140 bits/sec. What value of RTT in seconds (rounded off to the nearest integer) after which the TCP window scale option is needed?

$$\begin{aligned} & \text{Max}^m \text{ RTT (for No window Scale option needed)} \\ &= \frac{\text{Max}^m \text{ RWND without window Scale}}{\text{End-to-End Bandwidth}} = \frac{(2^{16} - 1) \text{ bytes}}{262140 \text{ bits/sec}} \\ &= \frac{(2^{16} - 1) * 2^3 \text{ bits}}{262140 \text{ bits/sec}} = 2 \text{ sec} \end{aligned}$$

Ans = 2

[NAT]



#Q. Consider 'window scale' option is opted during TCP connection establishment phase, the maximum possible window size with window scale option is $(2^{16} - 1) * 2^N$. What is the value of N?

$$\text{Scale_factor} = 2^{\text{shift_cnt}} \quad \left| \quad \begin{array}{l} \text{Max}^m \text{ value for shift_cnt} \\ \text{is 14 in TCP} \end{array} \right.$$

Max^m possible window size (RWND) after window scale

$$= (2^{16} - 1) * 2^{14} \text{ bytes}$$

$$= (2^{16} - 1) * 2^N \text{ bytes}$$

$$\boxed{N=14}$$

$$\boxed{\text{Ans}=14}$$

[MCQ]

#Q. Identify which name server(s) does not belong to DNS hierarchy ?

A

Root Name Server

B

Top Level Domain Name Server

C

Real Name Server

D

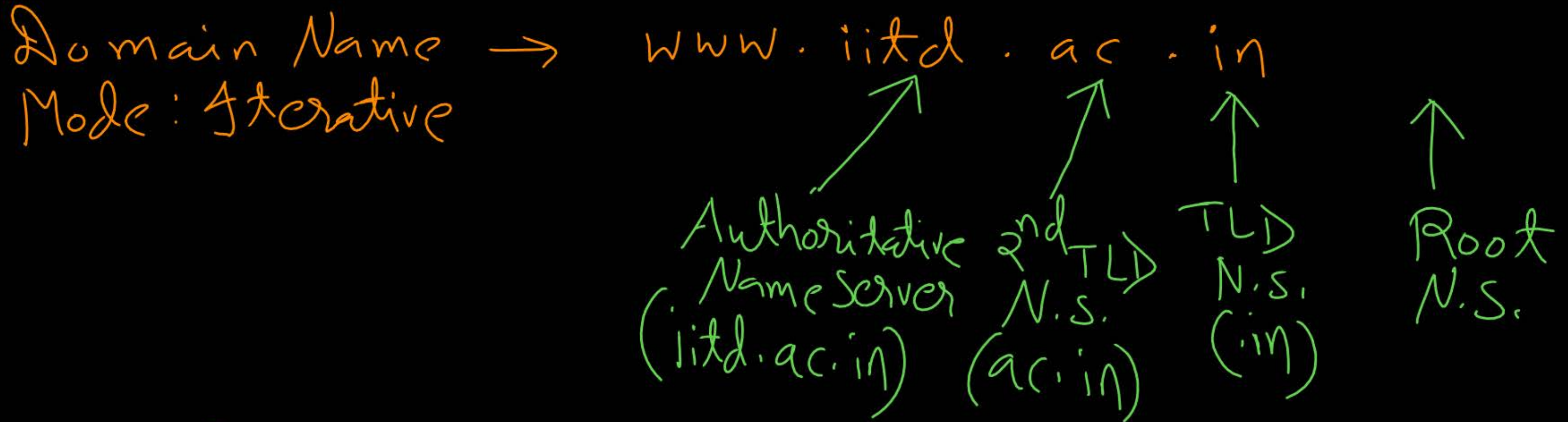
Authoritative Name Server

Ans: C

[NAT]



#Q. Consider the resolution of the domain name "www.iitd.ac.in" by a DNS resolver. Assume that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.



Ans = 4

[NAT]



#Q. Consider the resolution of the domain name "www.google.com" by a DNS resolver. Assume that iterative query mechanism is used in the resolution. Let RTT denote the round trip time between DNS resolver and DNS servers in the DNS hierarchy. The number of RTTs required in completely resolving the domain name is _____.

Domain Name = www.google.com
Mode: Iterative

Ans = 3 RTT

Ans = 3

↑
Authori-
-tative
N.S.

↑
TLD
N.S.
(.com)

↑
Root
N.S.

[NAT]



#Q. Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP. The web page contains base HTML file and four very small image objects. How many TCP connections required to load the web page completely?

Non-persistent HTTP

⇒ one TCP connection for base HTML File.

⇒ For every object, one TCP connection each

Total = 5 TCP connections

Ans = 5

[NAT]



#Q. Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP (no any parallel TCP connections). The web page contains base HTML file and four very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?

⇒ Non-persistent HTTP (No parallel TCP connection)

① 2 RTT for base HTML File.

② 2 RTT for each objects in sequential

Total = 10 RTT

Ans = 10

#Q. Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP that support parallel TCP connections. The web page contains base HTML file and four very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?

⇒ Non-persistent HTTP (parallel TCP connection)

① 2 RTT for base HTML File

② 2 RTT for four object in four parallel TCP connections

Total = 4 RTT

Ans = 4

[NAT]



#Q. Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP without pipeline. The web page contains base HTML file and four very small image objects. Assume base HTML file and all four image objects are placed in same web server. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?

⇒ Persistent HTTP (without pipeline)

- ① one RTT for TCP connection
- ② one RTT for base HTML File
- ③ one RTT for each object

Total = 6 RTT

Ans = 6

#Q. Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP with pipeline. The web page contains base HTML file and four very small image objects. Assume base HTML file and all four image objects are placed in same web server. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?

⇒ Persistent HTTP (with pipeline)

① one RTT for TCP connection

② one RTT for base HTML File

③ one RTT for all four objects in pipeline

Total = 3 RTT

Ans = 3

[MSQ]

#Q. Which is/are valid HTTP request method ?

☒ A GET

☒ B PUT

☒ C POST

☒ D HEAD

Ans: A, B, C & D

[MCQ]



#Q. Consider a web browser at client machine made a request for a web page to a web server. Identify the correct sequence in which the following protocols are used by the client host?

~~A~~

~~HTTP, TCP, DNS~~

B

DNS, TCP, HTTP

~~C~~

DNS, HTTP, TCP

~~D~~

~~TCP, DNS, HTTP~~

Ans: B



THANK - YOU